University Performance Data Warehousing & Visualization Capstone Project

Project Overview

This capstone project places you in the role of a Data Analyst for "Prestige University," a large multi-campus institution. The administration wants to move from siloed, departmental data to a unified warehouse to gain a holistic view of academic performance, student success, and operational efficiency. Your task is to design the data warehouse, populate it with realistic data, and create visualizations that provide actionable insights to faculty and administrators.

Business Scenario

Prestige University has multiple faculties, departments, and campuses. Key stakeholders include the Provost (concerned with overall academic performance), the Registrar (concerned with enrollment and records), and Department Chairs (concerned with faculty workload and student outcomes). They need answers to questions like:

- What are the trends in enrollment and graduation rates?
- How does student performance vary across departments, courses, and instructors?
- Are there demographic or socioeconomic factors influencing student success?
- How effectively are university resources (faculty, facilities) being utilized?

Project Requirements

1. Database Design (MySQL)

Create at least 7 tables following these schema patterns:

Central Fact Table:

- fact_course_enrollment (the core event: a student enrolling in and completing a course section)

Dimension Tables:

- dim_student (student demographics and academic info)
- dim_course (course details and hierarchy)
- dim_instructor (faculty information)
- dim_department (academic department info)
- dim_date (time dimension for semesters, years)
- dim_campus (campus location details)
- dim_faculty (parent faculty/school info, for snowflaking from Department)

2. Schema Implementation

Implement three different schema designs within the same project:

1. **Star Schema:** fact_course_enrollment connected directly to all dimensions (e.g., directly to dim_department).

2. **Snowflake Schema:** Normalize the hierarchy (e.g., fact_course_enrollment → dim_course → dim_department → dim_faculty).
3. **Flat Schema:** A single, denormalized table (vw_student_grades_flat) containing all information for specific quick reports.

3. Data Population

Create realistic data for the last three academic years (e.g., Fall 2021 - Spring 2024):

- 3 Campuses
- 5 Faculties (e.g., Arts, Science, Engineering, Business, Medicine)
- 15 Departments (3 per faculty)
- 50+ Courses
- 50+ Instructors
- 500+ Students
- 5,000+ Course Enrollment Records (well beyond the minimum 100 fact records)

4. Visualization Requirements

Create dashboards in each tool (**Excel, Power BI, and Tableau Public**) tailored to different stakeholders:

Provost's Executive Dashboard:

- University-wide enrollment trends over time
- Overall graduation rate and retention rate
- Average GPA by faculty and campus
- Demographic overview of the student body

Department Chair's Dashboard:

- Course enrollment heatmaps (popularity by semester)
- Average grade distribution for their department's courses
- Instructor workload analysis (number of sections taught)
- Student performance (DFW rate - % of D grades, Fails, Withdrawals) by course

Registrar's Operational Dashboard:

- Enrollment capacity vs. utilization
- Time-to-degree analysis
- Student credit hour generation by department
- Trends in major declarations

Project Implementation Guide

Phase 1: Database Design & Implementation

Design your tables with realistic attributes:

dim_student: student_id, first_name, last_name, date_of_birth, gender, ethnicity, enrollment_status, major_department_id, admission_date, expected_grad_date, campus_id

dim_course: course_id, course_code (e.g., 'MATH101'), course_name, course_description, credits, department_id

dim_instructor: instructor_id, first_name, last_name, title (Prof, Assoc Prof), department_id, hire_date

dim_department: department_id, department_name, department_code, faculty_id

dim_date: date_id, full_date, academic_year (e.g., '2022-2023'), academic_semester (Fall, Spring, Summer), academic_term (e.g., 'Fall 2022'), is_holiday, fiscal_period

dim_campus: campus_id, campus_name, location, chancellor

dim_faculty: faculty_id, faculty_name (e.g., "Faculty of Engineering"), dean

fact_course_enrollment: enrollment_id, student_id, course_id, instructor_id, semester_date_id (links to dim_date for the term), campus_id, final_grade (as a number, e.g., 4.0 for an A), grade_letter (A, B, C...), credits_earned, status (Completed, Withdrawn)

Phase 2: Data Generation and Population

1. **Hierarchy First:** Populate top-level dimensions first (dim_faculty -> dim_department -> dim_course).
2. **Realistic Data:**
    - Use realistic names (can be generated with online tools or Python's Faker library).
    - Create a realistic distribution of grades (typically a Bell Curve, slightly skewed higher).
    - Simulate student pathways: students should have a major, take courses primarily in their department, and have a logical sequence of semesters.
    - Include some realistic data issues: a few missing grades, a few withdrawn courses, some part-time instructors.

Phase 3: Visualization Development

Excel:

- Use PivotTables to quickly summarize grades by department and instructor.
- Create a dashboard with slicers for Academic Year, Semester, and Department.
- Use conditional formatting to highlight high DFW rates.

Power BI:

- Build a data model establishing the relationships between your fact and dimension tables.
- Create calculated measures using DAX for key metrics:
    - DFW Rate = DIVIDE([Number of DFWs], [Total Enrollments])
    - Total Student Credit Hours = SUM([credits_earned])

- Use bookmarks and buttons to create a navigational experience between the Provost, Chair, and Registrar views.

Tableau Public:

- Leverage the dim_date hierarchy to enable drill-down from Year -> Semester -> Term.
- Create calculated fields for cohorts (e.g., "Class of 2025").
- Use parameter controls to allow users to dynamically select which department or faculty to analyze.
- Use advanced visualizations like:
    - A funnel chart showing student retention from freshman to senior year.
    - A symbol map showing student distribution by home state (if you collect that data).

Deliverables

1. **SQL Scripts:** Complete .sql file that creates the schema, populates it with data, and includes sample analytical queries.
2. **Schema Diagram:** An ERD illustrating both the Star and Snowflake versions of your schema.
3. **Data Dictionary:** A document (PDF or Excel) defining each table and field.
4. **Visualization Files:** The three dashboard files (Excel, .pbix, Tableau Public link).
5. **Summary Report:** A 2-3 page PDF report summarizing your most critical findings for the university administration. What is one actionable insight you discovered?

Assessment Criteria

- **Database Design (25%):** Logical structure, appropriate use of keys, and implementation of all three schemas.
- **Data Quality & Realism (25%):** Data is coherent, follows academic rules, and is rich enough for analysis.
- **Analytical Depth (30%):** Visualizations answer complex business questions and reveal non-obvious insights.
- **Tool Proficiency (20%):** Effective and appropriate use of features in all three visualization platforms.

This project provides a fantastic portfolio piece that demonstrates your ability to handle complex, hierarchical data and translate it into insights for different business users.